

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting

findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Code of Conduct:

I N Mokshagna / 192472205 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N MOKSHA

Annexure A

Data Interpretation

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1 28 12000 65

P2 55 25000 40

P3 42 20000 75

P4 30 11000 70

P5 60 28000 35

SSE Values

K SSE

1 600

2 350

3 220

4 200

5 190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Answer:

Dataset: Hospital Patient Segmentation

Patient ID	Age	Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values:

K	SSE
1	600
2	350
3	220
4	200
5	190

Answer:

In WEKA, the SimpleKMeans algorithm is used to group patients based on their Medical Expense and Health Risk Score.

The given SSE values are:

- $K = 1 \rightarrow \text{SSE} = 600$
- $K = 2 \rightarrow \text{SSE} = 350$
- $K = 3 \rightarrow \text{SSE} = 220$
- $K = 4 \rightarrow \text{SSE} = 200$

- $K = 5 \rightarrow SSE = 190$

The SSE decreases rapidly from $K = 1$ to $K = 3$. After $K = 3$, the decrease is very small. Therefore, the elbow point occurs at $K = 3$.

Optimal number of clusters = 3

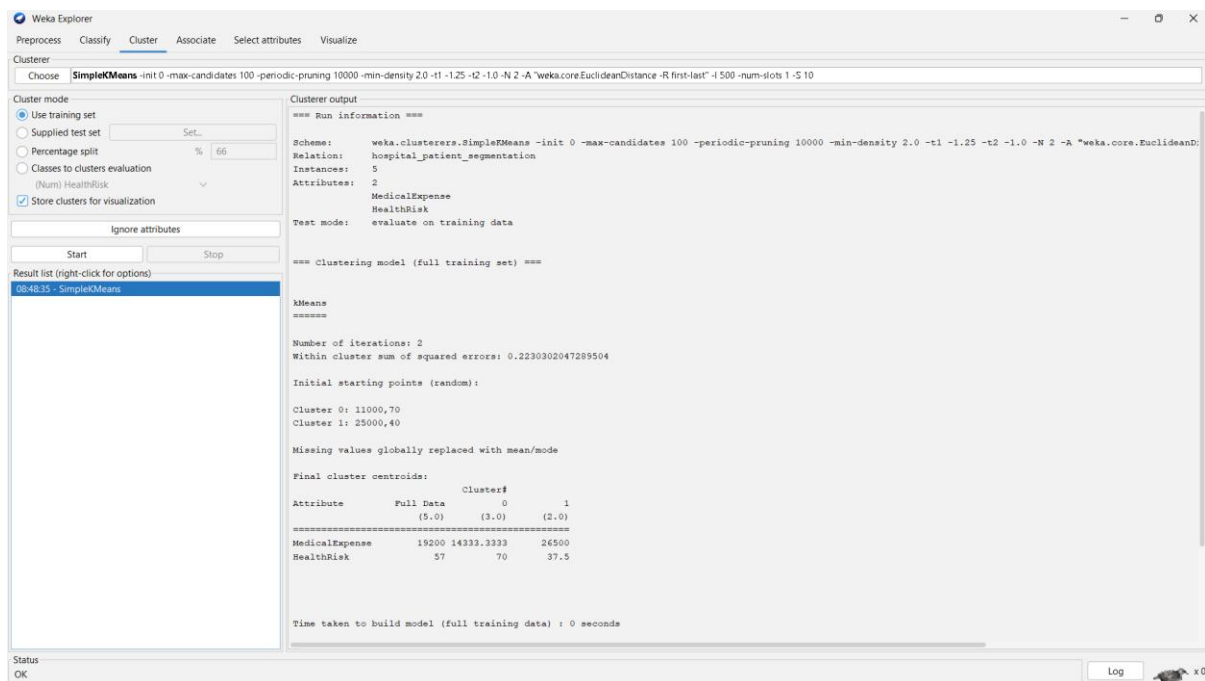
The three clusters can be used to segment patients according to their medical expenses and health risk. Patients with higher health risks can be given more attention and regular monitoring, while patients with higher medical expenses can be considered for cost-management programs.

Selecting 3 clusters helps the healthcare organization to:

1. Allocate healthcare resources effectively.
2. Identify high-risk patients.
3. Provide personalized healthcare services.
4. Plan preventive care and regular monitoring.
5. Manage medical costs efficiently.

Conclusion:

Using the Elbow Method, the optimal number of clusters is 3, because the SSE reduction becomes small after $K = 3$. This provides meaningful patient segmentation and supports better healthcare decision-making.



The screenshot shows the Weka Explorer interface with the SimpleKMeans model selected. The 'Clusterer' tab is active, displaying the command line and the resulting clustering output.

Clusterer output:

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Relation:    hospital_patient_segmentation
Instances:   5
Attributes:  2
              MedicalExpense
              HealthRisk
Test mode:   evaluate on training data

=== Clustering model (full training set) ===

KMeans
=====

Number of iterations: 2
Within cluster sum of squared errors: 0.2230302047289504

Initial starting points (random):

Cluster 0: 11000,70
Cluster 1: 25000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster#
              (5.0)      (3.0)      (2.0)
=====
MedicalExpense 19200 14333.3333 26500
HealthRisk     57      70      37.5

Time taken to build model (full training data) : 0 seconds
  
```

The 'Result list' on the left shows the selected model: 0b-48-35 - SimpleKMeans.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1 Electronics 55000

O2 Clothing 25000

O3 Electronics 57000

O4 Furniture 30000

O5 Clothing 27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Answer:

Algorithm: Hierarchical Clustering

In WEKA, the Hierarchical Clusterer algorithm is used to group the orders based on their Order Value.

The order values are:

Order ID	Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

Cluster Formation

From the hierarchical clustering dendrogram:

- O1 and O3 are grouped together because their order values are very close (₹55,000 and ₹57,000).
- O2 and O5 are grouped together because their values are close (₹25,000 and ₹27,000).
- O4 (₹30,000) is closer to the lower-value orders than the high-value electronics orders.

Therefore, a suitable choice is 2 clusters:

- Cluster 1: O1, O3 → High-value orders
- Cluster 2: O2, O4, O5 → Low/medium-value orders

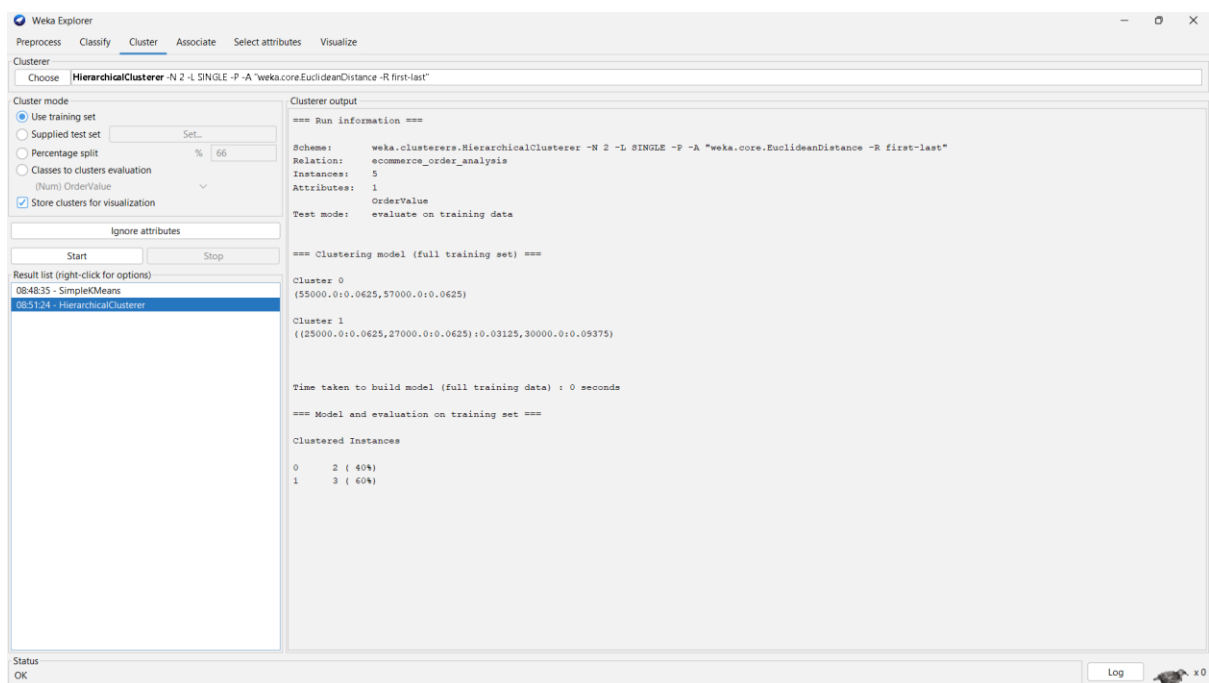
Significance

The selected clusters help the retailer understand customer purchasing behavior:

1. High-value cluster: Represents customers making expensive purchases, mainly electronics.
2. Low/medium-value cluster: Represents customers making relatively lower-value purchases.
3. The retailer can provide premium offers and loyalty programs to high-value customers.
4. Discounts and promotional offers can be used to increase spending in the lower-value group.
5. Clustering helps in customer segmentation, targeted marketing, and sales planning.

Conclusion

The hierarchical clustering dendrogram indicates that 2 clusters are appropriate. These clusters separate high-value orders from low/medium-value orders, helping the retailer understand purchasing patterns and make better marketing decisions.



Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1 120 18 2000

U2 60 5 5000

U3 140 22 1500

U4 50 4 6000

U5 130 20 1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their

activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Answer:

Algorithm: Hierarchical Agglomerative Clustering (HAC)

HAC is used to group mobile app users based on their **daily usage time, login frequency, and in-app purchase amount**.

Cluster Formation

Based on the user behavior, the users can be grouped into two main clusters:

- **Cluster 1: Highly Active Users** → U1, U3, U5
These users have high daily usage time and high login frequency. Their in-app purchases are relatively lower.
- **Cluster 2: Less Active but High-Spending Users** → U2, U4
These users have low daily usage time and login frequency but make higher in-app purchases.

Probability-Based Membership

In **soft clustering**, a user can belong to more than one cluster with different membership probabilities.

For example:

User	Active User Cluster	High-Spending Cluster
U1	0.90	0.10
U2	0.20	0.80
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

For example, **U2 has 80% membership in the High-Spending cluster and 20% in the Active User cluster**. This shows that users do not always belong completely to only one group.

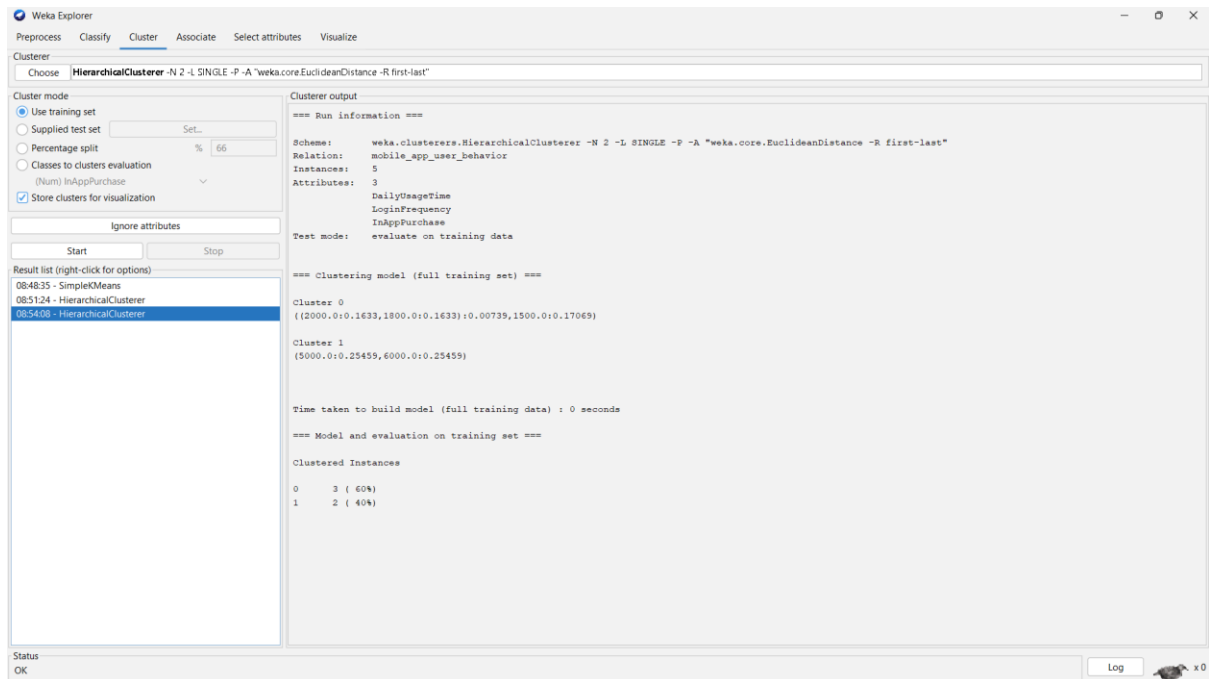
Advantages of Soft Clustering

1. **Flexible:** A user can belong to multiple clusters.
2. **Better behavior analysis:** It represents mixed customer behavior more realistically.
3. **Identifies borderline users:** Users between two groups can be easily identified.
4. **Better marketing:** Different offers can be provided according to membership probabilities.

5. **Improved decision-making:** Helps companies understand complex customer behavior.

Conclusion

Soft clustering provides more flexibility than hard clustering because users can have **different membership probabilities in multiple clusters**. This is useful for mobile applications because user behavior can change and may not fit into a single fixed category.



Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1 300 8 85

F2 500 4 70

F3 400 6 80

F4 280 9 90

F5 520 3 65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Answer:

Q4. Agricultural Crop Monitoring – Clustering Before and After Normalization

Algorithm: K-Means Clustering

The farm dataset contains three attributes: **Rainfall, Soil Fertility Score, and Crop Yield**. Clustering is performed first on the original data and then after normalization.

Before Normalization

The attributes have different scales:

- Rainfall: **280–520 mm**
- Soil Fertility: **3–9**
- Crop Yield: **65–90 tons**

Rainfall has much larger numerical values than the other attributes. Therefore, K-Means distance calculations are mainly influenced by **rainfall**.

This can result in clusters that are based mostly on rainfall rather than considering soil fertility and crop yield equally.

After Normalization

Normalization converts the attributes into a similar scale, usually **0 to 1**.

After normalization:

- Rainfall, soil fertility, and crop yield have comparable influence.
- No single attribute dominates the distance calculation.
- Clusters are formed based on the overall similarity of farms.
- The clustering results become more balanced and meaningful.

Difference in Clustering

Before Normalization	After Normalization
Rainfall dominates	All attributes have similar influence
Distance is affected by large values	Distance is more balanced
Clusters may be biased	Clusters are more meaningful
Less reliable analysis	More reliable analysis

Importance of Normalization

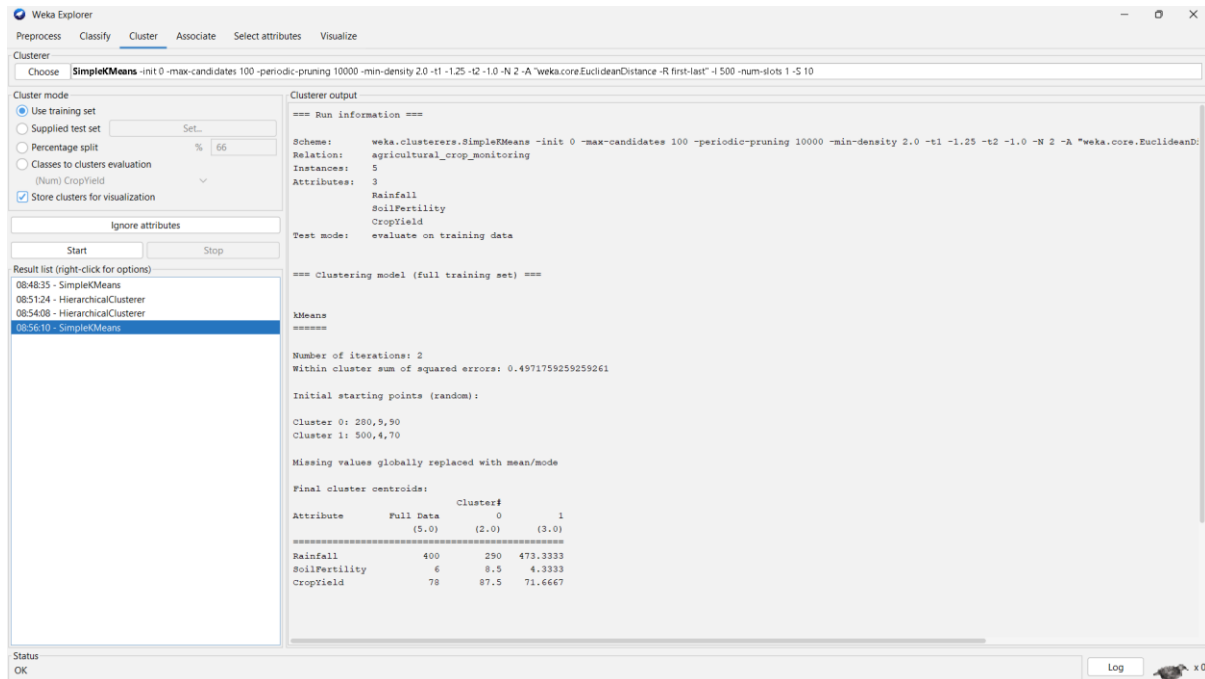
Normalization improves agricultural data analysis because it:

1. Prevents large-scale attributes from dominating.
2. Gives equal importance to different features.
3. Produces more accurate and meaningful clusters.
4. Improves the performance of distance-based algorithms like K-Means.

5. Helps researchers identify farms with similar agricultural conditions.
6. Supports better decisions about irrigation, fertilizers, and crop planning.

Conclusion

Normalization significantly influences cluster formation by putting all attributes on a comparable scale. After normalization, the clustering results are more balanced, reliable, and useful for agricultural decision-making.



5. Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1 88 92 84

E2 62 75 68

E3 78 88 72

E4 55 65 58

E5 93 96 90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores.

Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answer:

Algorithm: K-Means Clustering

Number of Clusters (K) **Silhouette Score**

$K = 2$	0.48
$K = 3$	0.68
$K = 4$	0.52

Answer:

The silhouette score measures how well employees are grouped into clusters. A higher score indicates better-defined and more separated clusters.

- $K = 2 \rightarrow 0.48$: Moderate clustering
- $K = 3 \rightarrow 0.68$: Best clustering
- $K = 4 \rightarrow 0.52$: Moderate clustering

Therefore, the most suitable number of clusters is $K = 3$, because it has the highest silhouette score of 0.68.

The three clusters can represent employees with high, medium, and low performance.

Conclusion:

$K = 3$ is the best model because it provides well-separated and meaningful employee groups. It helps the company identify high performers, employees needing improvement, and provide suitable training or rewards.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) SkillAssessment

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 08:48:35 - SimpleKMeans
- 08:51:24 - HierarchicalClusterer
- 08:54:08 - HierarchicalClusterer
- 08:56:10 - SimpleKMeans
- 08:58:04 - SimpleKMeans

Cluster output

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last -I 500 -num-slots 1 -S 10

Relation: employee_performance_evaluation

Instances: 5

Attributes: 3

ProductivityScore

Attendance

SkillAssessment

Test mode: evaluate on training data

=== Clustering model (full training set) ===

KMeans

====

Number of iterations: 3

Within cluster sum of squared errors: 0.39597925805094436

Initial starting points (random):

Cluster 0: 55,65,58

Cluster 1: 62,75,68

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	
		0	1
	(5.0)	(2.0)	(3.0)
ProductivityScore	75.2	58.5	86.3333
Attendance	83.2	70	92
SkillAssessment	74.4	63	82

Status

OK

Log

x 0